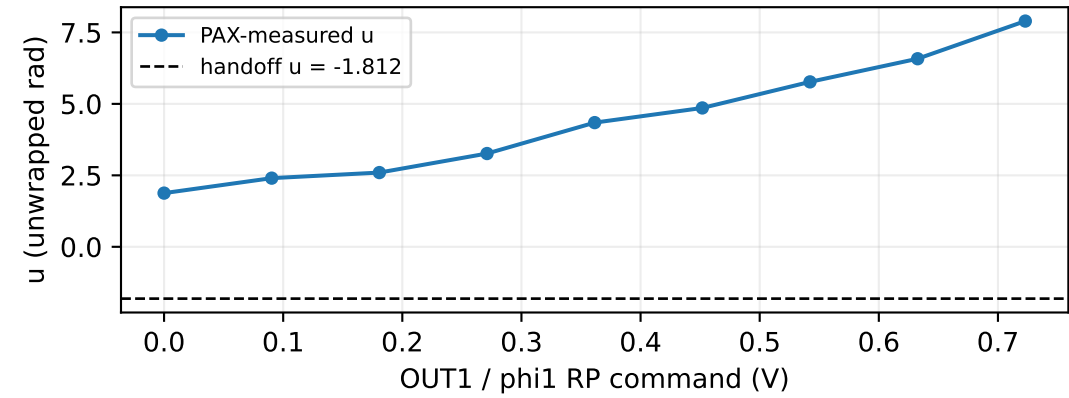
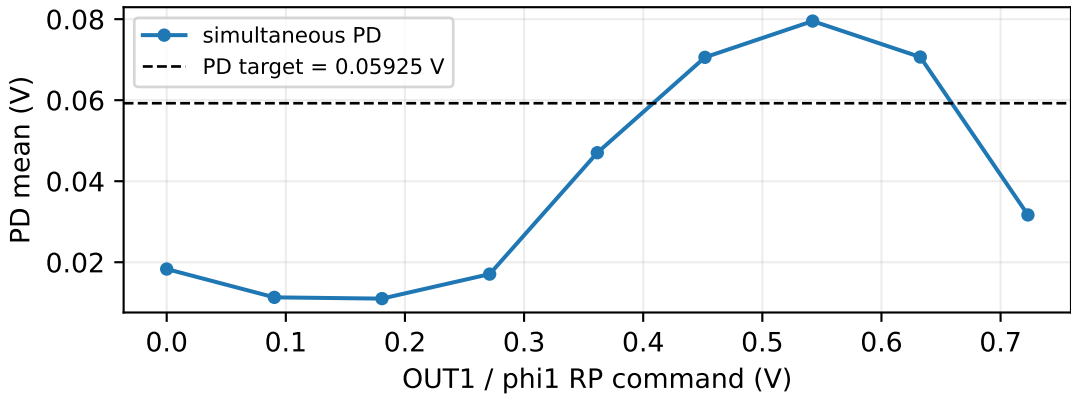


Phi1 PD FPGA lock — 012334_phi1-pd-lock-test_phi1-pd-lock-6

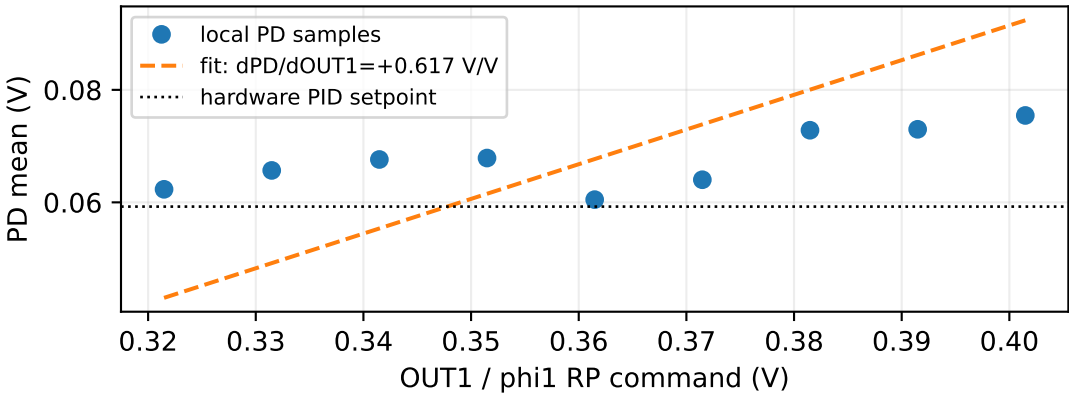
PAX-assisted $\phi 1$ rough map



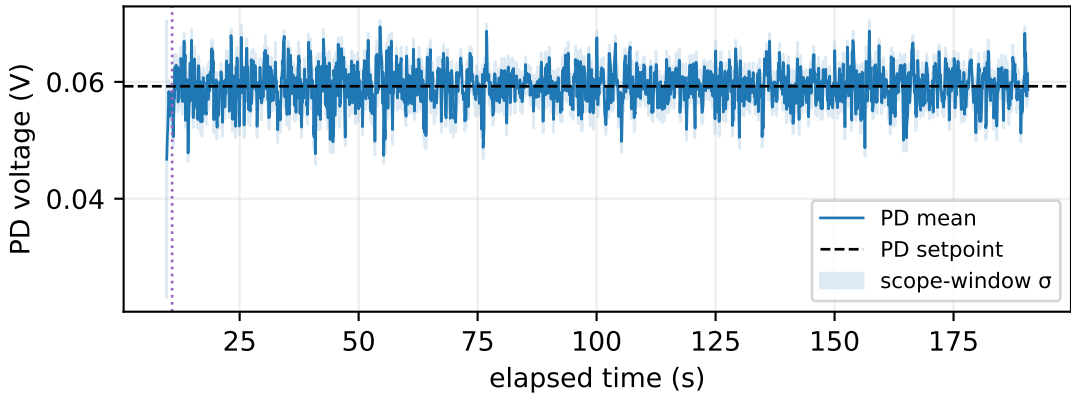
PD value associated with rough u map



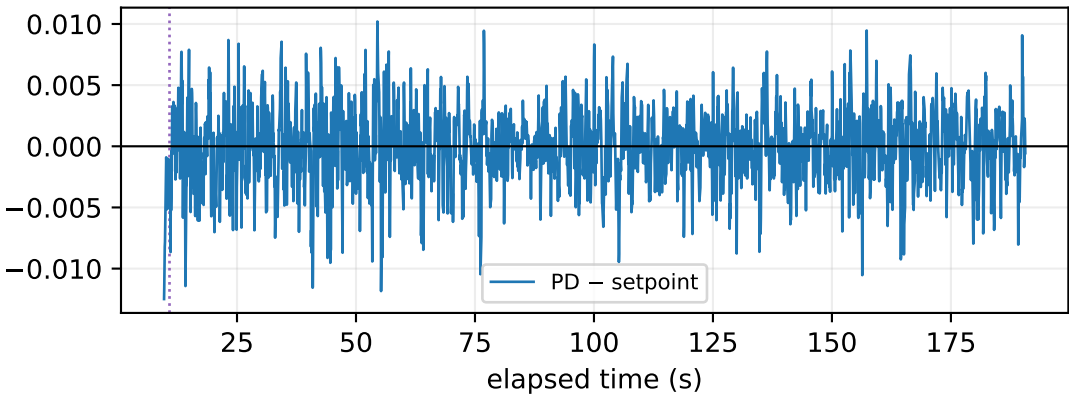
Automatic local PD calibration



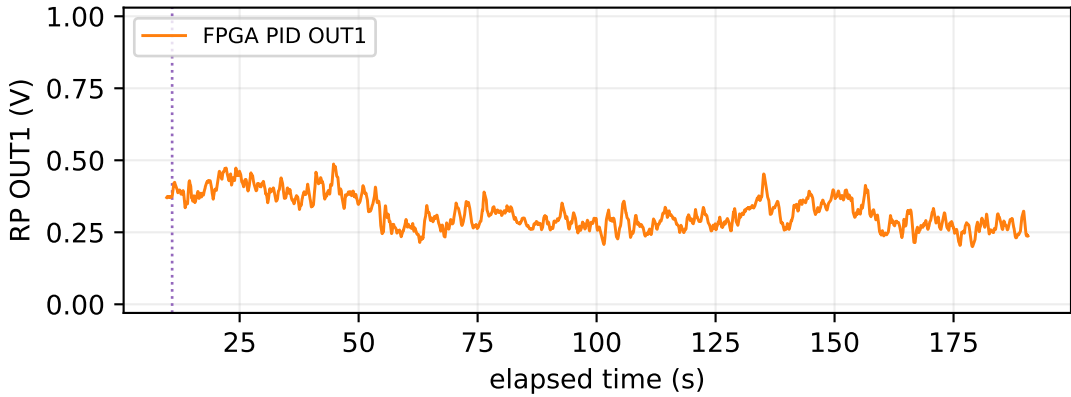
FPGA PD lock



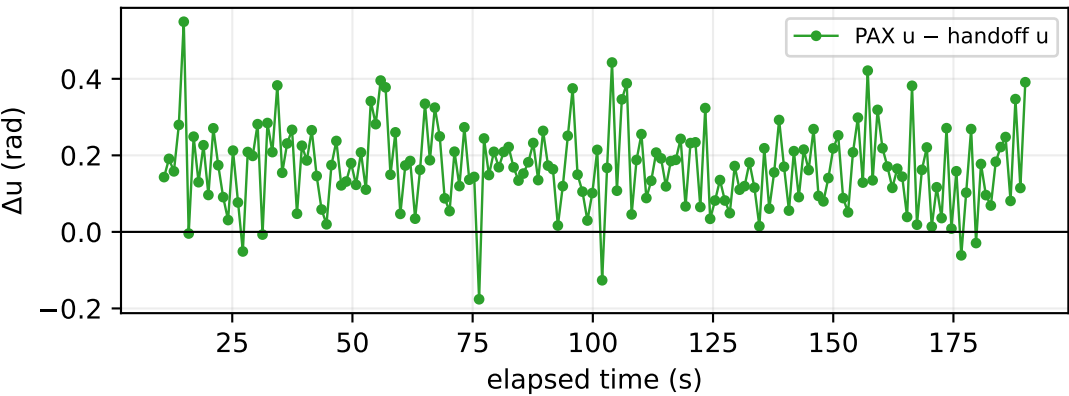
PD locking error



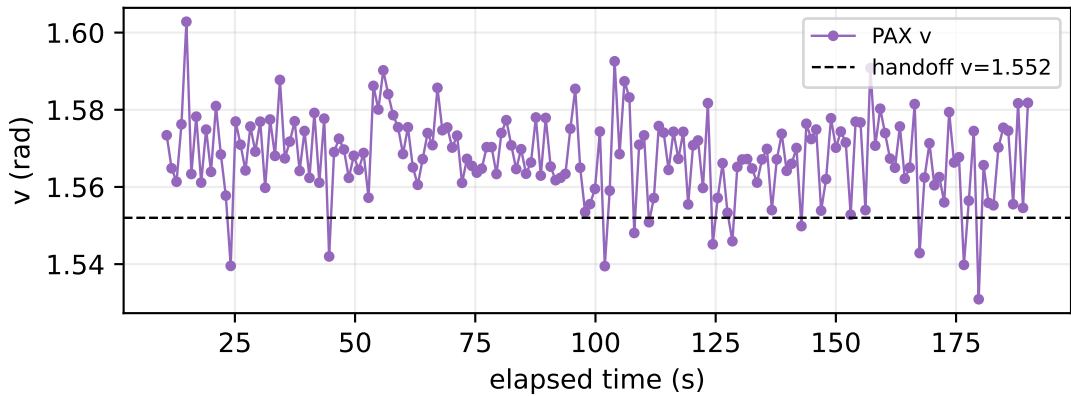
Continuous FPGA actuator output



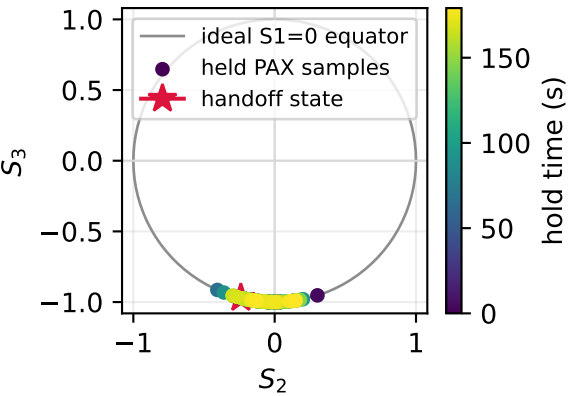
Independent PAX azimuth validation



PAX polar-angle cross-check



Held lock trajectory on the Poincare equator



Interpretation

- Tight cluster near the red star: PD lock preserves the selected u branch.
- Motion around the circle: residual $\phi 1/u$ drift or branch movement.
- Radial motion: polarization is leaving the ideal equator.

PAX handoff target $u = -1.8124$ rad PD setpoint = 0.05925 V local slope = +0.6168 V/V
FPGA PID: P = -0.276, I = -5 Hz Duration = 181.0 s median |PD error| = 0.00186 V (late 0.00193 V) OUT1 range = 0.200...0.487 V PAX median $|\Delta u| = 0.169$ rad (176 snapshots)